STAT 516 - Homework 3 Answer Key

1. **(12 points)** In a company, 12% of employees in their first five years earn \$100,000 or more, and 24% of employees beyond their first five years earn \$100,000 or more. Overall, 15.6% of employees earn \$100,000 or more. Determine the proportion of employees in their first five years.

   Let p be the proportion of employees in their first five years.
   0.12p + 0.24(1-p) = 0.156
   0.24 - 0.12p = 0.156
   0.12p = 0.084, so p = 0.70.

2. **(13 points)** A disease affects 2% of a population. A medical test is used to detect the disease.

If a person has the disease, the test gives a positive result with probability 90%. If a person does not have the disease, the test gives a positive result with probability 0.1%.

A randomly selected person tests positive. What is the probability that this person actually has the disease?

Let D be the event that the person has the disease, and let T be the event that the test is positive.

$$P(D|T) = \frac{P(T \mid D)P(D)}{[\mathrm{P(T \mid D)P(D)} + \mathrm{P(T} \mid D^c)\mathrm{P}(D^c)]}$$
$$= \frac{0.90(0.02)}{0.90(0.02) + 0.001(0.98)}$$
$$= \frac{0.018}{0.01898}$$
$$= 0.9484$$

3. **(10 points)** A student takes an examination consisting of true-false questions. If the student knows the answer to a question, then the student answers correctly. Otherwise, the student guesses the answer. If the student does not know the answer, the probability of guessing correctly is 0.5. For a given question, the probability that the student knows the answer, given that the student answered correctly, is 0.75. Find the probability that the student knows the answer to the question.

   Let K be the event that the student knows the answer, and let C be the event that the student answers correctly.
   If the student does not know the answer, the probability of guessing correctly is 0.5.
   Let P(K) = p. Then P(C) = p + 0.5(1 - p).
   Given P(K | C) = 0.75, we have p / [p + 0.5(1 - p)] = 0.75.
   p / (0.5 + 0.5p) = 0.75.
   p = 0.375 + 0.375p, so 0.625p = 0.375.
   p = 0.60.

4. **(16 points)** Find the appropriate value of c for each of the following functions so that each defines a probability mass function on the positive integers 1,2,3, ...

(a) $p_X(x) = c\, 2^{-x}$

(b) $p_X(x) = c\, \frac{2^x}{x!}$

(a)

Consider the case $P_X(x) = c\,2^{-x}$.

Notice that the probability measure has to be non-negative and should add up to 1. The non-negative condition implies $c > 0$.

Using the normalisation condition:

$$1 = \sum_{i=1}^{\infty} P_X(i)$$

$$1 = c\sum_{i=1}^{\infty} 2^{-i}$$

$$1 = (2-1)c$$

$$\implies c = 1$$

Note that the above series is just geometric series of $\sum_{i=0}^{\infty} 0.5^x = 1/(1-0.5)$ except the first term 1. Hence we subtract 1.

(b)

Consider the case $P_X(x) = c\,2^x/x!$.

Again $c > 0$ and using the normalisation condition:

$$1 = \sum_{i=1}^{\infty} P_X(i)$$

$$1 = c\sum_{i=1}^{\infty} 2^i/i!$$

$$1 = (e^2-1)c$$

$$\implies c = \frac{1}{e^2-1}$$

Note that the above series is just Taylor series of e^x at $x = 2$ except the first term 1. Hence we subtract 1.

5. (10 points) A continuous random variable X has density function

$$f(x)=\begin{cases} \frac{1}{(1+x)^2} & x \geq 0 \\ 0 & otherwise \end{cases}$$

Calculate $P(1 < X < 4)$.

$$P(1 < X < 4) = \int_1^4 \frac{1}{(1+x)^2} dx$$
$$= -\frac{1}{1+x}\Big|_1^4$$
$$= -\frac{1}{5} + \frac{1}{2}$$
$$= \frac{3}{10}$$

6. **(12 points)** For a random variable X, you are given the PDF

$$f(x)=\begin{cases} cx^3 & 0 \leq x \leq 2 \\ 0 & otherwise \end{cases}$$

Calculate $P(0.5 \leq X \leq 1.5)$.

First find c by requiring the density to integrate to 1.

$$1 = \int_0^2 cx^3\, dx = \frac{cx^4}{4}\Big|_0^2 = 4c$$

Thus $c = \frac{1}{4}$.

$$P(0.5 \leq X \leq 1.5) = \int_{0.5}^{1.5} \frac{1}{4} x^3\, dx$$

$= \frac{x^4}{4(4)}\Big|_{0.5}^{1.5} = \frac{1}{16}$ (1.5^4 - 0.5^4)= 5/16=0.3125.

7. **(12 points)** The distribution of the size of claims paid under an insurance policy has PDF

$$f(x)=\begin{cases} cx^a & 0 < x < 6 \\ 0 & otherwise \end{cases}$$

where a > 0 and c > 0. For a randomly selected claim, the probability that the size of the claim is less than 4.5 is 0.50.

Calculate the probability that the size of the claim is greater than 5.

Since $f(x) = cx^a$ on $0 < x < 6$, the CDF has the form $F(x) = (\frac{x}{6})^{a+1}$

Given $F(4.5) = 0.50$, we have $(\frac{4.5}{6})^{a+1} = 0.50.$

So $a + 1 = \frac{ln(0.50)}{ln(0.75)}$

$$P(X > 5) = 1 - F(5) = 1 - (\frac{5}{6})^{a+1}$$

$$= 1 - (5/6)\text{^}[\frac{ln(0.50)}{ln(0.75)}] \approx 0.3549.$$

8. **(15 points)** The distribution of the amount of time, in hours, for a staff meeting has PDF

$$f(x)=\begin{cases} c(x-1)^5(2-x) & 1 < x < 2 \\ 0 & otherwise \end{cases}$$

Calculate the probability that a staff meeting lasts longer than 1.6 hours.

Let y = x-1. Then 0 < y < 1, and the density is proportional to y^5 (1-y).
The normalizing constant satisfies c integral from 0 to 1 of y^5 (1-y) dy = 1.
$1 = \int_0^1 cy^5 (1-y)\, dy = c(\frac{1}{6}-\frac{1}{7})\Big|_0^1 = c\frac{1}{42}$, so c = 42.

We need P(X > 1.6) = P(Y > 0.6).

$$P(Y > 0.6) = \int_{0.6}^{1} y^5 (1-y)dy.$$

$$= 42(\frac{1}{6}y^6 - \frac{1}{7}y^7)\Big|_{0.6}^{1}$$

$$=1 - [7(0.6^6) - 6(0.6^7)].$$

$$\approx 0.8414.$$